
FROM-SCRATCH TRANSFORMER • ~65M PARAMETERS

Architecting a Small Language Model (SLM)

Sanskrit-to-English Translation of the Bhagavad Gita

1. Abstract & Objectives

This document specifies the architecture for designing, training, and deploying a Small Language Model (SLM) dedicated to translating Sanskrit text — with a primary focus on the Bhagavad Gita — into English. The design follows a strict “from-scratch” methodology: a compact, custom Transformer trained without reliance on massive pre-trained weights, prioritising computational efficiency, precise handling of ancient Sanskrit syntax, and context-aware philosophical translation.

Large Language Models are multilingual but computationally heavy and prone to hallucinating cultural or philosophical nuance in ancient texts. An SLM in the 50–150 million parameter range offers a focused, domain-specific alternative. The objective is a sequence-to-sequence model that translates Sanskrit slokas (verses) into English while preserving the theological and philosophical accuracy of the source.

2. Dataset Strategy & Acquisition

The Bhagavad Gita alone contains only 700 verses — far too few for training a model from scratch. A two-stage data strategy resolves this by first building general Sanskrit–English competence, then fine-tuning on the Gita itself.

Phase 1 — Base Language Acquisition

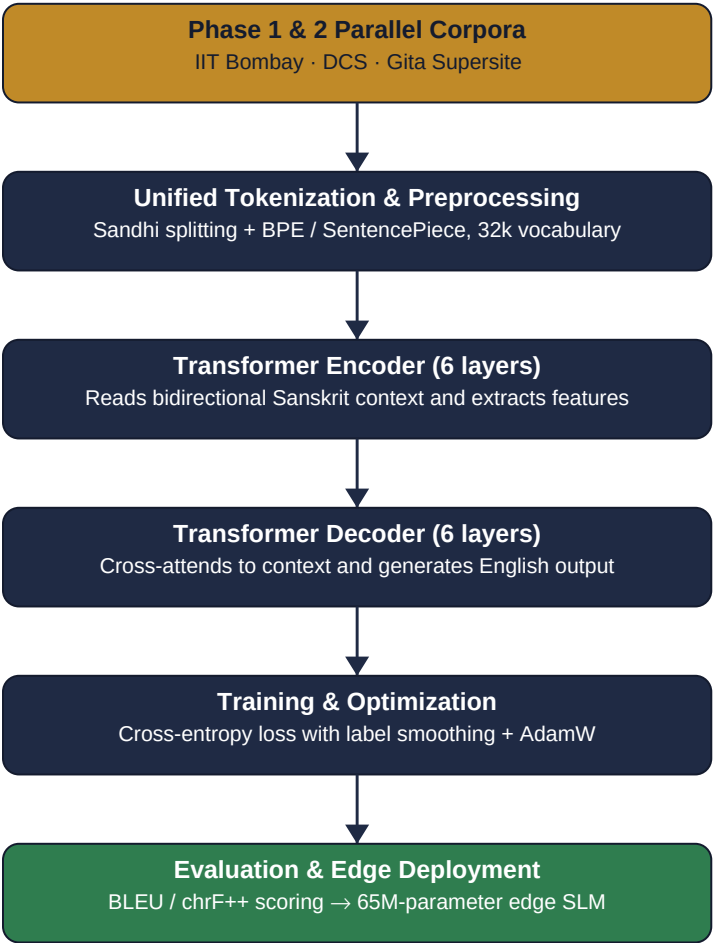
- **IIT Bombay English–Hindi/Sanskrit Corpus** — established Sanskrit–English sentence alignments.
- **JNU Sanskrit Corpora** — grammatical structures, root words (Dhatus), and English equivalents.
- **Digital Corpus of Sanskrit (DCS)** — lexical resources mapping Sanskrit terms to English meanings.

Phase 2 — Domain-Specific Fine-Tuning

- **Gita Supersite (IIT Kanpur)** — Devanagari verses, IAST transliteration, and multiple English translations.
- **Hugging Face Datasets** — structured JSON parallel corpora such as *sanskrit_english_parallel*.

3. System Architecture & Flow

For a pure translation task at small scale, a Transformer Encoder–Decoder is more efficient than a decoder-only (GPT-style) model. The encoder reads the full Sanskrit verse to capture bidirectional context; the decoder generates the English translation sequentially, cross-attending to that context.



SLM Blueprint — “Gita-Translate-Small”

Total parameters: ~65 million, optimised for edge deployment and fast inference.

Component	Specification	Rationale
Model Type	Transformer Encoder–Decoder	Ideal for Seq2Seq tasks; captures bidirectional context of complex Sanskrit grammar.
Layers	6 Encoder / 6 Decoder	Sufficient depth for linguistic mapping without overfitting a small corpus.
Attention Heads	8	Attends to distinct grammatical rules, prefixes, and suffixes simultaneously.
Embedding Dimension	512	Enough dimensional space for semantic meaning while keeping memory footprint low.
Feed-Forward Dim	2048	Standard 4x expansion ratio for internal non-linear transformations.

4. Tokenization Strategy

Sanskrit is highly agglutinative: words fuse together through Sandhi (phonetic junctions), so a standard English tokenizer will fail on it.

Aspect	Detail
Algorithm	Byte-Pair Encoding (BPE) or Unigram-based SentencePiece
Vocabulary Size	32,000 tokens, shared between Sanskrit and English
Preprocessing	Text standardised to IAST or preserved in Devanagari; a single tokenizer trained on the combined corpus lets the model bridge sub-word elements across both languages

5. Training Methodology (From Scratch)

Because the architecture contains no pre-trained weights, the training pipeline is rigorously structured around two components.

Objective Function

Cross-Entropy Loss with Label Smoothing (typically 0.1). Label smoothing prevents overconfidence in translations — important given that a single Sanskrit word can carry multiple philosophical interpretations.

Optimization & Hardware Profile

An AdamW optimizer is used with a learning-rate schedule featuring a warm-up phase followed by linear decay. At 65M parameters, the model can be trained from scratch on a single consumer-grade GPU (e.g. RTX 3090/4090) within a matter of days.

6. Evaluation Metrics

Metric	Purpose
BLEU Score	Precise n-gram overlap between model output and the reference human translation.
chrF++	Character n-gram metric that correlates highly with human judgment; well suited to morphologically rich input.
Semantic Preservation	Human-in-the-loop review of tone, philosophical accuracy, and readability, since automated metrics miss theological nuance.

7. Conclusion

Constraining the scope to a 65-million-parameter Encoder–Decoder architecture yields a highly targeted, computationally inexpensive solution for Sanskrit-to-English translation. Public datasets provide base training, while fine-tuning on the Bhagavad Gita ensures the output is grammatically correct and philosophically resonant — a fast, efficient model suited to academic, spiritual, and edge-device applications.